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Year Level & Section:	**4-D**	**Student #**	**202311177**

Activity #1 - Finals
Database Integration

PART 1: TRACING OUTPUT

```
insert.setInt(1, 103);
insert.setString(2, "Ana Cruz");
insert.setString(3, "BSIT");
insert.setInt(4, 4);

int rows = insert.executeUpdate();
```

Question 1: What record will be inserted? 103, Ana Cruz, BSIT, 4
Question 2: What will the value of rows be? 1
Question 3: After the INSERT operation, how many records should exist? 3
Initial:
101 Maria Santos
102 Juan Reyes

After INSERT:
101 Maria Santos
102 Juan Reyes
103 Ana Cruz

Trace the UPDATE Operation
Consider:

```
update.setString(1, "BSCS");
update.setInt(2, 103);

int rows = update.executeUpdate();
```

Question 4: Which student will be updated? Ana Cruz
Question 5: What is the student's course before the update? BSIT
Question 6: What is the student's course after the update? BSCS
Question 7: What is the value of rows? 1

Trace the DELETE Operation
Consider:

```
delete.setInt(1, 103);

int rows =
 delete.executeUpdate();
```

Question 8: Which record will be deleted? 103 / Ana Cruz
Question 9: How many records remain after deletion? 2
Question 10: What are the remaining records? [{101, “Maria Santos”}, {102, “Juan Reyes”}]

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DEPARTMENT OF COMPUTER STUDIES
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

PART II: TRACING OUTPUT

Problem:
Assume the database contains:
201 | Carlo Mendoza | BSIT | 4
202 | Lea Garcia | BSCS | 3

The program executes:

```
String sql =
 "SELECT name FROM students WHERE id = ?";

PreparedStatement ps =
 connection.prepareStatement(sql);

ps.setInt(1, 202);

ResultSet rs =
 ps.executeQuery();

if (rs.next()) {
 System.out.println(
 "Student: " +
 rs.getString("name")
 );
}
```

What is the output?: Lea Garcia

Program Tracing

```
String sql =
 "SELECT course FROM students WHERE id = ?";

PreparedStatement ps =
 connection.prepareStatement(sql);

ps.setInt(1, 999);

ResultSet rs =
 ps.executeQuery();

if (rs.next()) {
 System.out.println(
 rs.getString("course")
 );
} else {
 System.out.println(
 "Student not found."
 );
}
```

Question: If student 999 does not exist, what is the output?

- Student not found